

Labwork 3

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September 30, 2026

1 Objective

For this labwork, I need to convert the RGB image to Grayscale image using both CPU and GPU. But instead of using 1D block, I choose to improve by using 2D blocks. After that, I calculate the time to measure the speedup

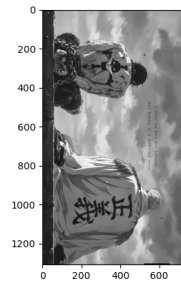
2 Python pseudo-code

- Read the image using `plt.imread`
- Reshape to 2D array
- Apply the grayscale for the image using CPU
- Apply the grayscale for the image using GPU
- Compare the different of time between each blocks

3 Results



(a) CPU result



(b) GPU result

Figure 1: Grayscale conversion results using the CPU and GPU.

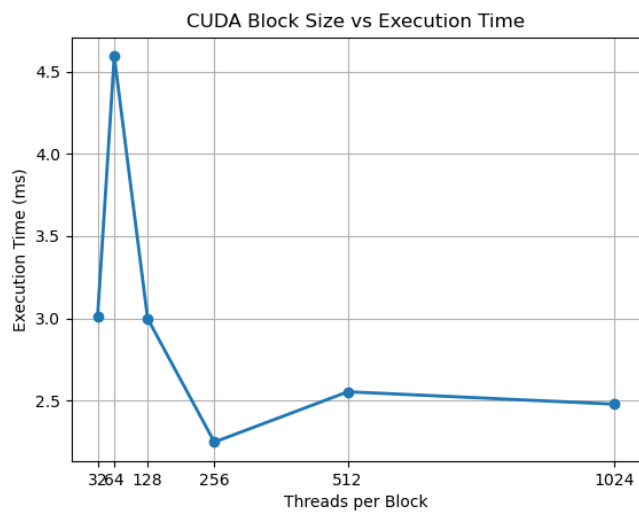


Figure 2: CUDA block size versus GPU execution time.